



PUCP
Departamento
Académico de Ingeniería

IA
PUCP

Inteligencia Artificial para el reconocimiento de lenguaje de señas



Summer Camp 2025

Joe Huamani

Joe Huamani

- 2018 - Becado para estudios en Japón
- 2019 - Pregrado: ingeniería informática PUCP
- 2020 - Desarrollador de aplicaciones móviles
- 2022 - Diccionario Lengua de Señas Peruana
- ML engineer, AI researcher, MLOps, Web Developer
- Actualmente:
 - AI2050 (AI researcher)
 - Micro Master en “Estadística y ciencia de datos” en MIT



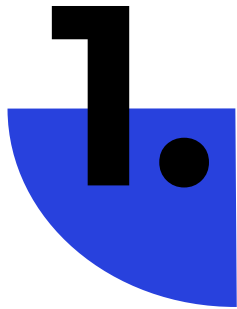
Agenda

1. Introducción básica del trabajo con Lengua de seña
2. PERUSIL: a Sign Language Annotation Framework
3. Machine Learning used to Measure Sign Language Similarity
4. Impact of Pose Estimation on the Sign Language Recognition Task
5. Peruvian Sign Language Dictionary
6. Comparing Incremental Learning Approaches for a Growing Sign Language Dictionary
7. AI for Sign Language Data Interpolation
8. Iconicity in Sign Language

Agenda

0%

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Introducción básica del trabajo con Lengua de seña

Lengua de señas

¿Qué es?



Lengua de señas

¿Qué es?

Lengua visual con estructura y gramática propia

Involucra movimiento de las manos, pose y gestos



Lengua de señas

¿Es una lengua universal?



Lengua de señas

¿Es una lengua universal?

Varía según la región o país

Según la Federación Mundial de Sordos (World Federation of the Deaf), existen:

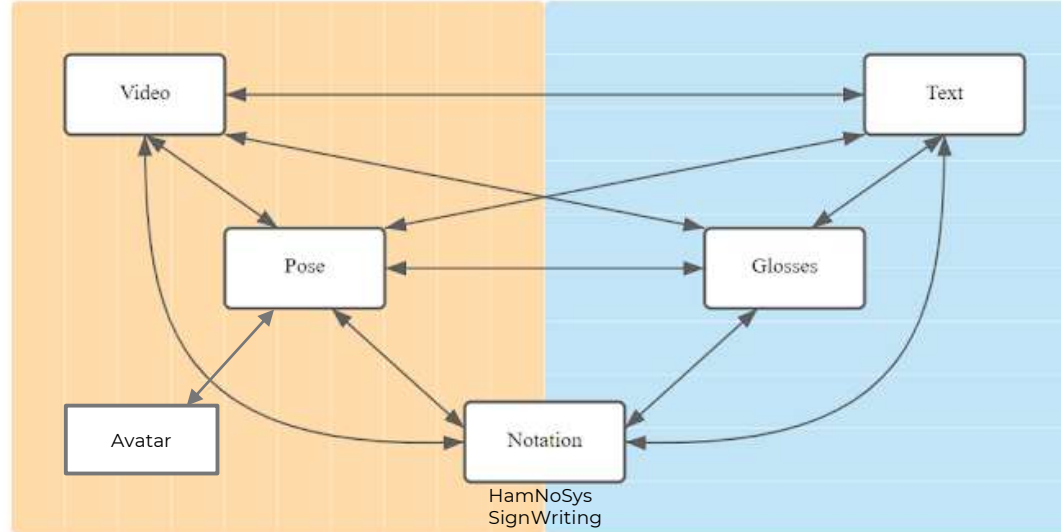
- 70 millones de personas sordas en el mundo
- Más de 200 lenguas de señas



Sign Language Processing

Language Agnostic Tasks

Language Specific Tasks



Glosa: Convención de la lengua de seña que ayuda describirla de forma escrita

Notación: sistema que se adopta para expresar conceptos

In total, there are 20 tasks conceptually defined by this graph, with varying amounts of previous research. Every path between two nodes might or might not be valid, depending on how lossy the tasks in the path are.

Sistemas de notación

Glosa: ventaja en la anotación inicial (palabra parecida a la lengua oral o escrita)

HamNoSys y SignWriting: notación y sistema que implica mayor conocimiento

Gloss

HamNoSys

SignWriting



La meta más ambiciosa



Voz/Texto

Avatar (SL)



Gallaudet
University
CREST Network

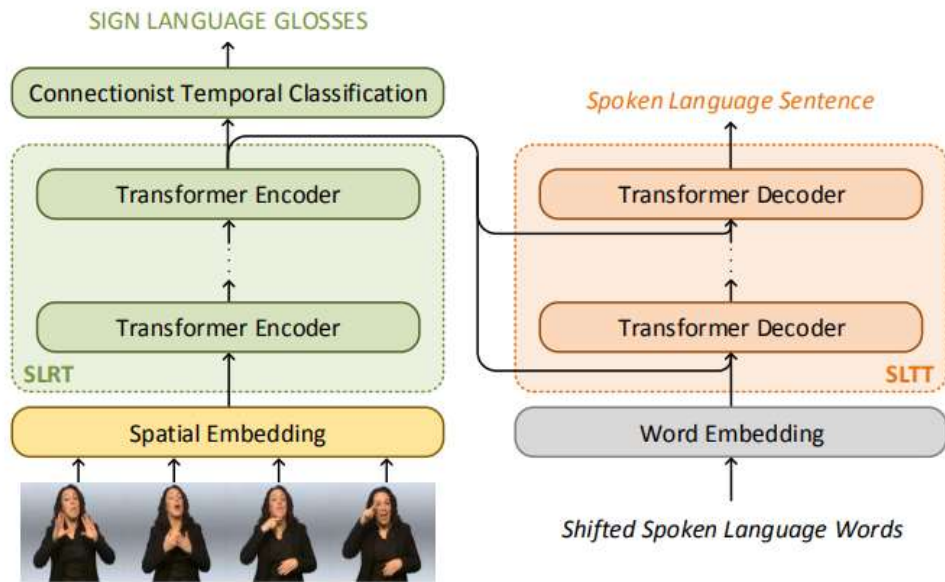


Voz/Texto

Video/pose



Video to Text



[Camgoz, et al CVPR 2020]

Pose to Avatar

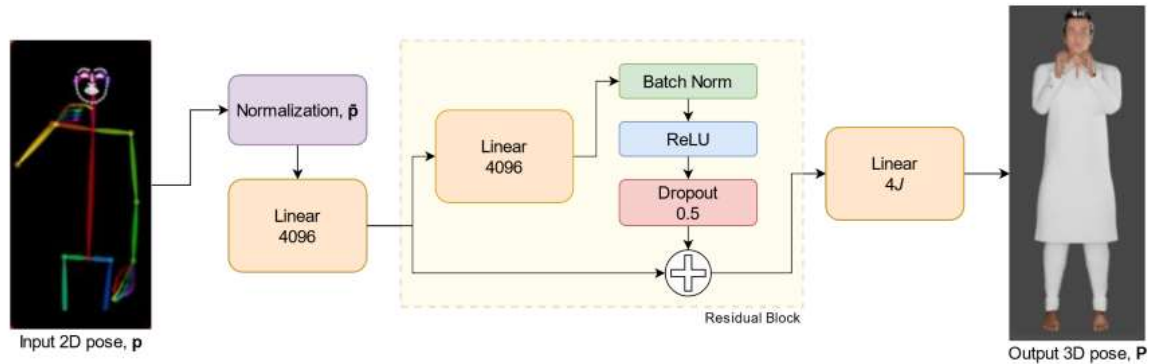


Figure 2: Architecture of the model network.

[Krishna, et al ICCV Workshop 2021]

Introducción básica del trabajo con Lengua de seña

- 
1. Concepto
 2. Tipos de procesamiento
 3. Notaciones
 4. Metas

Agenda

6.25%

1. Introducción básica del trabajo con Lengua de seña
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2.

PERUSIL: a Sign Language Annotation Framework

Anotación de glosas

ELAN
herramienta
para la
anotación

Genera un archivo EAF
(estructura similar a html)

Usar la librería “pympi-ling”
de Python

Capas o tiers para
anotar glosa
individuales y
traducciones/oraciones



Anotación de glosas

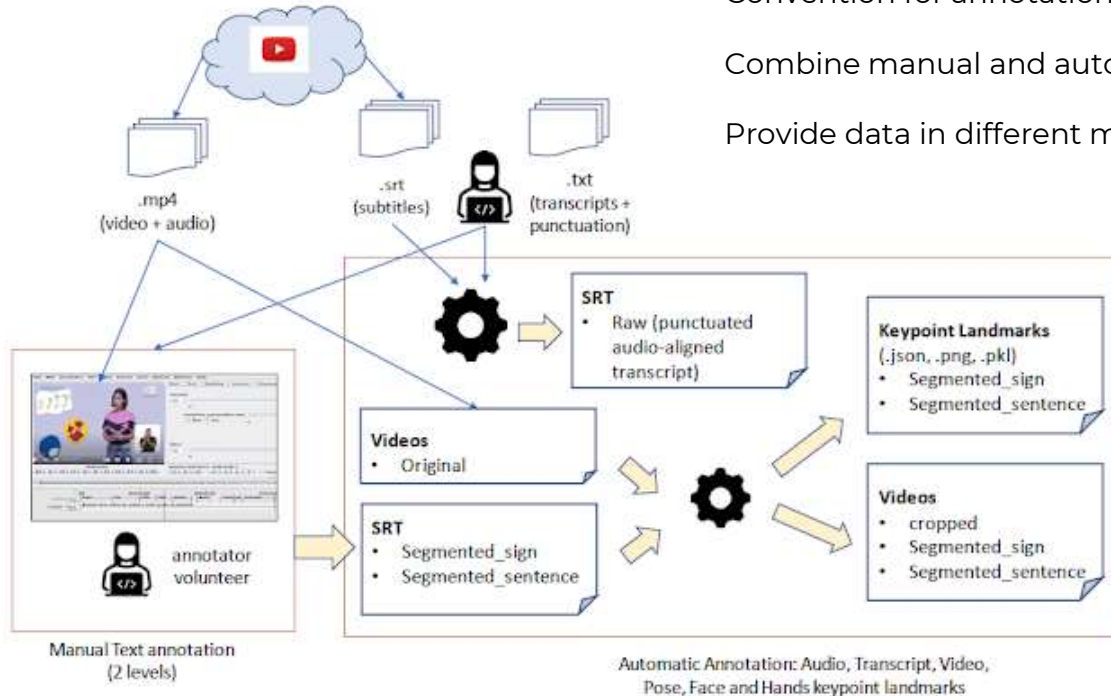
ELAN
herramienta
para la
anotación

La anotación de una
lengua debe ser
hecha por nativos en
esa lengua

Capas o tiers para
anotar glosa
individuales y
traducciones/oraciones



Framework PERUSIL



Convention for annotation

Combine manual and automated process

Provide data in different modalities



[Paper](#)

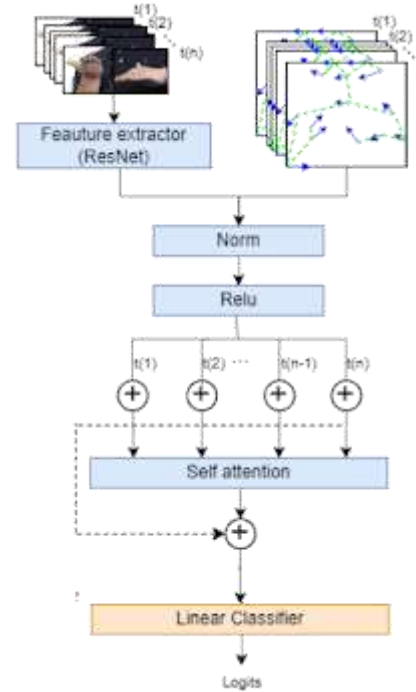
SL Recognition Model

5 glosses with more than 10 instances

Train:AEC

Test: PUCP-DGI

Feature extractor	Sequence Length	Learning Rate	Stride	F1 micro AEC (%)	F1 micro PUCP-DGI (%)
rn18	10	1.0E-04	1	80.3 ± 2.6	20.2 ± 6
rn18	10	1.0E-05	1	59.1 ± 4.5	46.4 ± 4.9
rn18	10	1.0E-06	1	30.3 ± 5.2	49.6 ± 2.7
rn18	10	1.0E-04	2	78.8 ± 2.6	20.8 ± 7.9
rn18	10	1.0E-05	2	53 ± 6.9	48.8 ± 7.3
rn18	10	1.0E-06	2	30.3 ± 6.9	51.8 ± 3.6
rn18	15	1.0E-04	1	78.8 ± 2.6	22.2 ± 4.5
rn18	15	1.0E-05	1	57.6 ± 6.9	50.6 ± 9.8
rn18	15	1.0E-06	1	31.8 ± 4.5	49.8 ± 4.1
rn18	15	1.0E-04	2	74.2 ± 5.2	30 ± 11.5
rn18	15	1.0E-05	2	50 ± 4.5	50.4 ± 9.5
rn18	15	1.0E-06	2	31.8 ± 0	51 ± 1.5



[Code](#)

PERUSIL: a Sign Language Annotation Framework

- 
1. Anotación
 2. Framework
 3. Modelo
 4. Resultados

Agenda

18.75%

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3.

Machine Learning used to Measure Sign Language Similarity

[Das and Bejarano, 2022]

Sign Language Workshop LREC

Similaridad en Lengua de Señas



“Trabajar” en ASL
(American Sign Language)

“Trabajar” en LSP
(Lengua de Señas Peruana)

Similaridad en Lengua de Señas



“Trabajar” en ASL
(American Sign Language)



“Trabajar” en LSP
(Lengua de Señas Peruana)

Similaridad en Lengua de Señas

Un trabajo muy laborioso

Se evalúan 4 características:

Movimiento

Orientación

Forma

Ubicación



“Trabajar” en ASL



“Trabajar” en LSP

Otros ejemplos

Misma forma

Diferente ubicación



Sobrino vs sobrina

Otros ejemplos

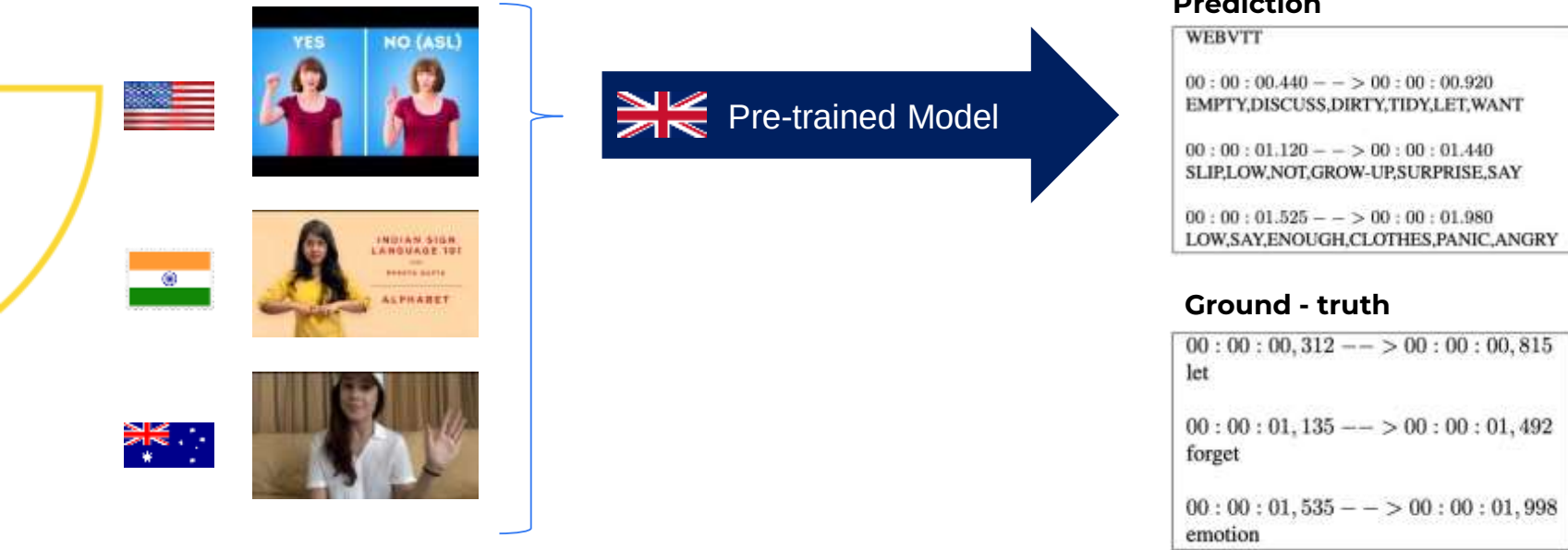
Misma forma y ubicación

Diferente movimiento



Sobrino vs terco

Similaridad en Lengua de Señas



input: sign language "sentence"

Similaridad en Lengua de Señas



Pre-trained Model

Interpret Accuracy
as overlap or
similarity

input: sign language "sentence"

Prediction

WEBVTT

00 : 00 : 00.440 -- > 00 : 00 : 00.920
EMPTY,DISCUSS,DIRTY,TIDY,LET,WANT

00 : 00 : 01.120 -- > 00 : 00 : 01.440
SLIP,LOW,NOT,GROW-UP,SURPRISE,SAY

00 : 00 : 01.525 -- > 00 : 00 : 01.980
LOW,SAY,ENOUGH,CLOTHES,PANIC,ANGRY

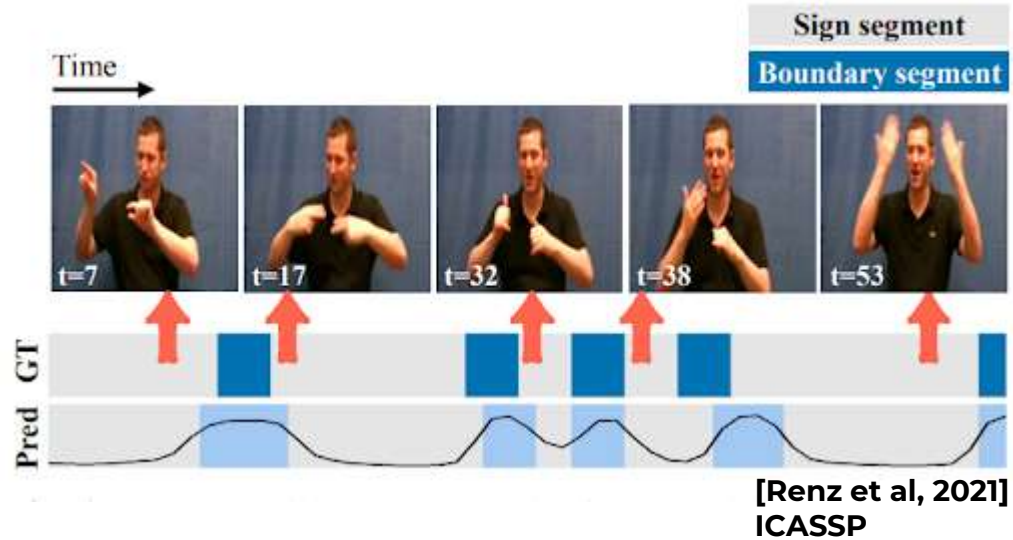
Ground - truth

00 : 00 : 00,312 -- > 00 : 00 : 00,815
let

00 : 00 : 01,135 -- > 00 : 00 : 01,492
forget

00 : 00 : 01,535 -- > 00 : 00 : 01,998
emotion

Similaridad en Lengua de Señas



Model:

- Segmentation
- Classification

Evaluación de Traducción (Machine Translation Evaluation)

- Evaluación Humana vs Evaluación Automática
- Traducción:
 - BLEU
 - METEOR
 - Word Error Rate

Metrics

- Exact Match
- Synonym Match
- Double Synonym

Sign-Word List

- Swadesh (100)
- Woodward (excluding pointing signs)

Prediction

```
WEBVTT

00:00:00.440 --> 00:00:00.920
EMPTY,DISCUSS,DIRTY,TIDY,LET,WANT

00:00:01.120 --> 00:00:01.440
SLIP,LOW,NOT,GROW-UP,SURPRISE,SAY

00:00:01.525 --> 00:00:01.980
LOW,SAY,ENOUGH,CLOTHES,PANIC,ANGRY
```

Ground - truth

```
00:00:00,312 --> 00:00:00,815
let

00:00:01,135 --> 00:00:01,492
forget

00:00:01,535 --> 00:00:01,998
emotion
```

Swadesh 100 original final list [\[edit \]](#)

Swadesh's final list, published in 1971,^[5] contains 100 terms. Explanations of the 1955. Note that only this original sequence clarifies the correct meaning which is specification here added).

- | | |
|-------------------------------|--|
| 1. I (Pers.Pron. 1.Sg.) | 27. bark (of tree) |
| 2. you (2.sgl 1952 thou & ye) | 28. skin (1952: person's) |
| 3. we (1955: inclusive) | 29. flesh (1952 meat, flesh) |
| 4. this | 30. blood |
| 5. that | 31. bone |
| 6. who? ("?" not 1971) | 32. grease (1952: fat, organic substance) |
| 7. what? ("?" not 1971) | 33. egg |
| 8. not | 34. horn (of bull etc., not 1952) ^T |
| 9. all (of a number) | 35. tail |
| 10. many | |

Wikipedia

Results

Sign Language	Woodward Similarity (in %)			Swadesh Similarity (in %)			Classic Similarity (in %)
	Exact Match	Synonym Match	Double Synonyms	Exact Match	Synonym Match	Double Synonyms	
ASL 	28.57	33.33	47.62	13.95	23.26	39.53	25
Auslan 	23.72	50	57.69	34.62	46.15	48.72	77
ISL 	0	9.09	9.09	0	0	0	7

[Das and Bejarano et al, 2022]
Sign Language Workshop LREC

Results

Auslan Sign Language	MATCH_SYNG (in %)		
	Swadesh Similarity	Woodward Similarity	Classic Similarity
Northern 1	39.29	25	
Northern 2	45	28.57	
Northern 3	27.59	23.53	
Total of 3 files	36.36	25	77
Melbourne	55.17	75	
Sidney	68	61.11	
Total of 2 files	63.29	68.42	77
Table3: Disaggregated analysis of Australian dialects.			



[Paper](#)

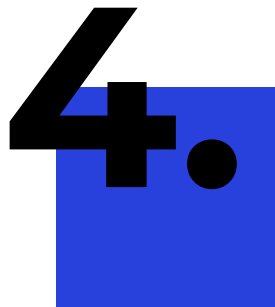
Machine Learning used to Measure Sign Language Similarity

- 
1. Concepto y ejemplos
 2. Modelo y sus usos
 3. Evaluación
 4. Resultados
- 

Agenda

45.83%

1. Introducción básica del trabajo con Lengua de seña
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Impact of Pose Estimation on the Sign Language Recognition Task

[Lazo et al, 2022]

LXAI Workshop - ECCV

Motivation

- OpenPose was the most used library for sign language research
- Measure the quality of Pose Estimation Models
- Found the granularity needed in the pose for Sign Language Recognition models

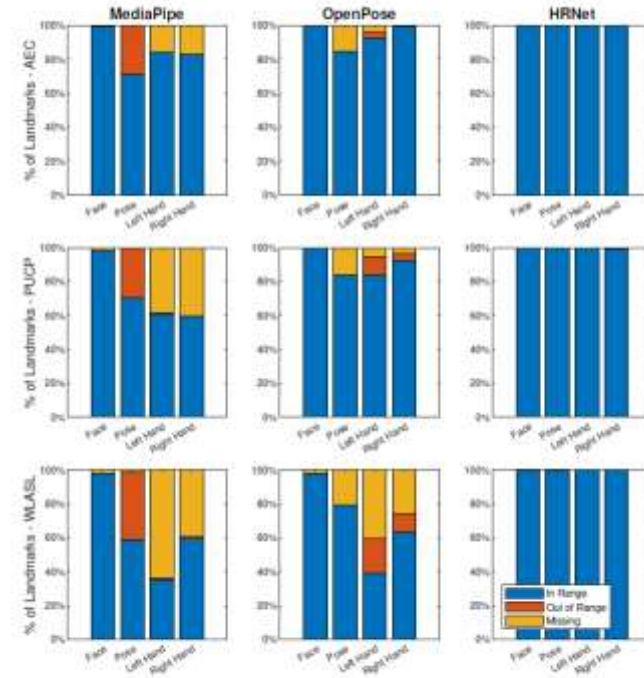


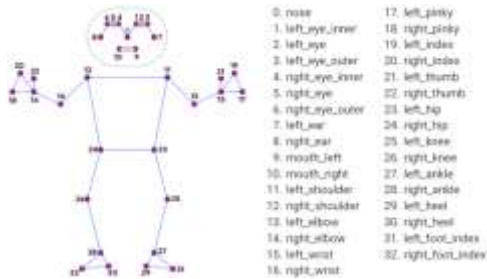
Fig. 2: Categories of quality landmarks in body section Left Hand, Right Hand and Face

Pose Estimation Libraries

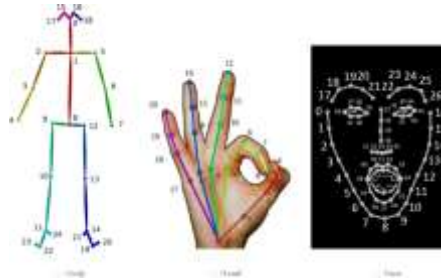
Different granularity, models and datasets used for training

29 vs 71 points

MediaPipe



OpenPose



HRNet

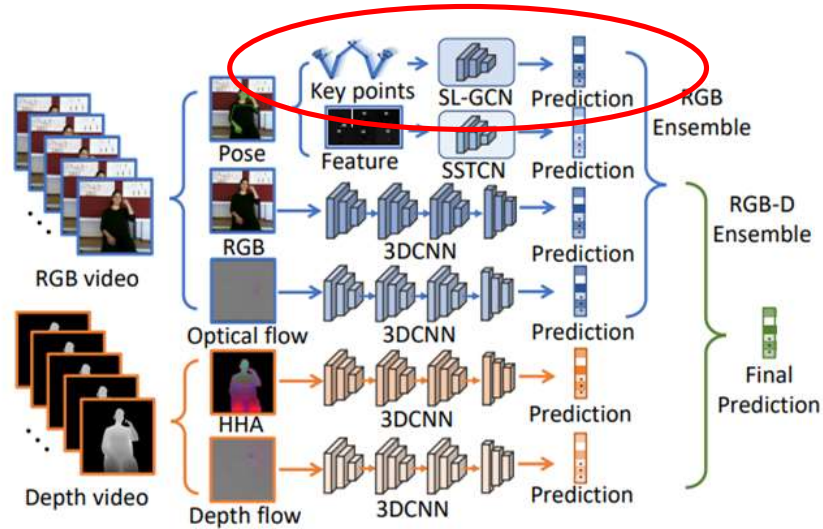


Sign Language Recognition Models

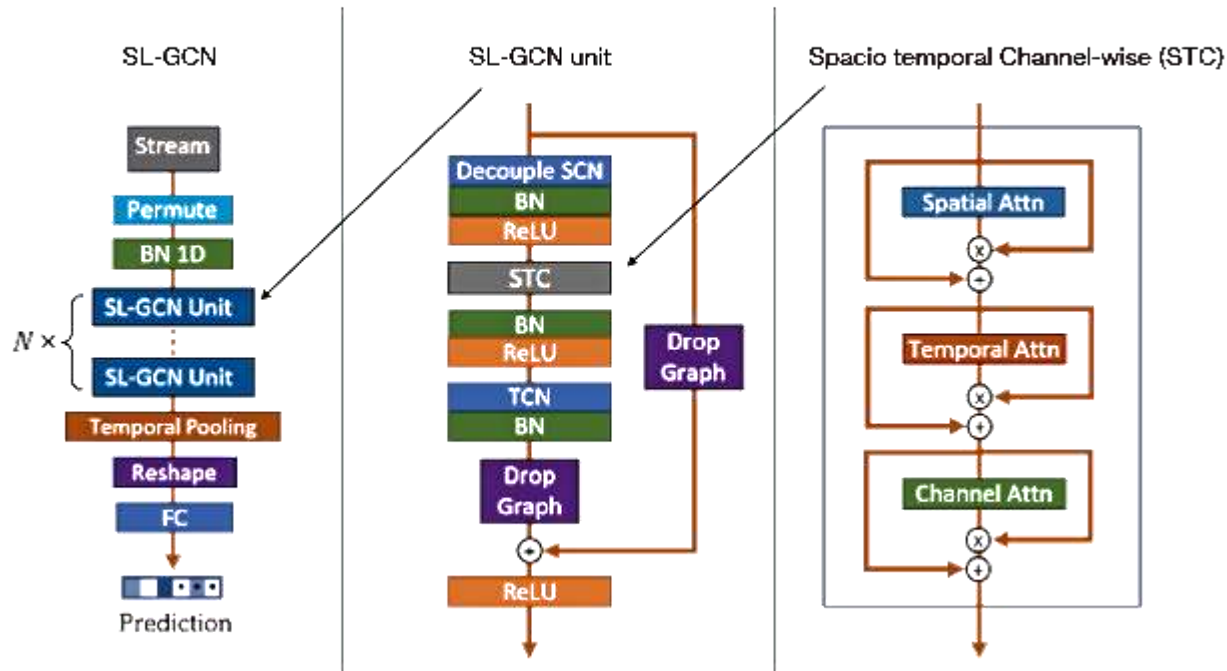
- Graph-based: SL-GCN
- Transformer-based: Spoter



Graph based: SL-GCN

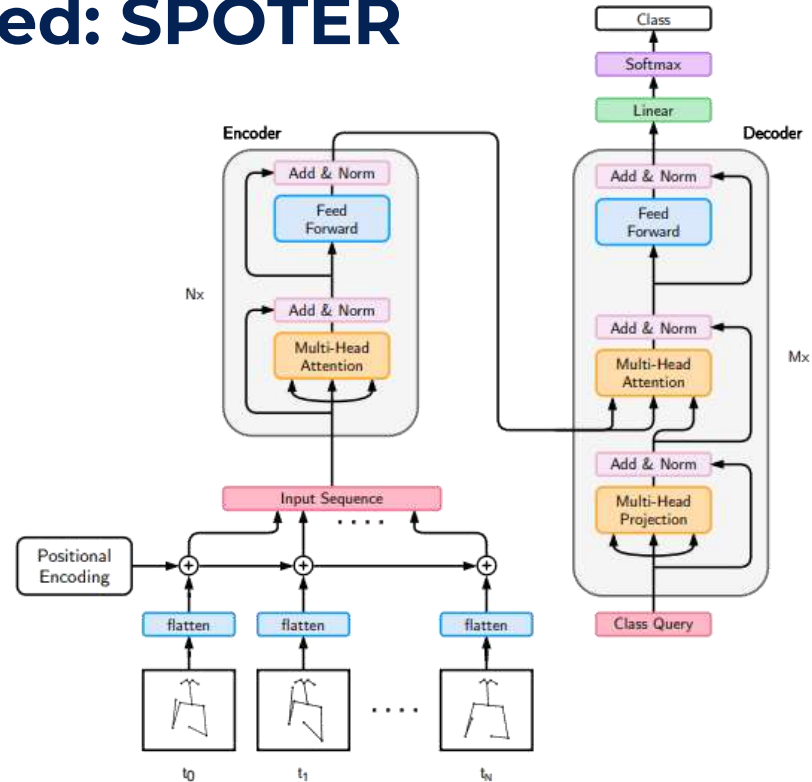


Graph based: SL-GCN



[Code](#)

Transformer based: SPOTER



[Code](#)

[Bohacek and Hruz, 2021]
WACV

Results

SLR Model	Library	Top-1					
		AEC		PUCP		WLASL	
		29	71	29	71	29	71
Spoter	MediaPipe	0.728	0.7452	0.4481	0.4973	0.6237	0.7073
	OpenPose	0.612	0.6306	0.5683	0.5464	0.4739	0.5679
	HRNet	0.696	0.6943	0.4973	0.4918	0.3671	0.4755
SL-GCN	MediaPipe	0.732	0.738	0.562	0.475	0.759	0.724
	OpenPose	0.700	0.687	0.5683	0.557	0.710	0.630
	HRNet	0.751	0.707	0.5737	0.535	0.590	0.543



[Paper](#)

TABLE I: Spoter [5] and SL-GCN [14] Top-1 results for groups of keypoints: 29 and 71

Impact of Pose Estimation on the Sign Language Recognition Task

1. Motivación
2. Librerías
3. Modelos
4. Resultados



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58.33%

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5.

Peruvian Sign Language Dictionary

Trabajo Interdisciplinario



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Miguel Rodriguez
Líder de Lingüística



Joe Huamaní
Ingeniero de Aprendizaje de Máquina



Sabina Oporto
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Carlos Vasquez
Ingeniero de Datos
@cvasquezroque



Cristian Lazo
ML Research Engineer
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Francisco Cerna
Apoyo de Anotación y protocolo de pruebas
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César Ramos
Apoyo de Anotación



Juan Villamonte
Consultor LSP



Alexandra Arnaiz
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Julio Mendoza
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Jenny Vega
Asesora en AWS



Daniel Carbajal
Asesor en AWS y de Desarrollo de Software



Rossy Latorraca
Soporte de Interfaz Web



Corrado Doly
Soporte de la Nube



STRATLABS

Centrarse en el trabajo PARA y CON la comunidad sord

Why Sign-Language Gloves Don't Help Deaf People

Wearable technologies that claim to translate ASL overlook the intricacies of the language, as well as the needs of signers.

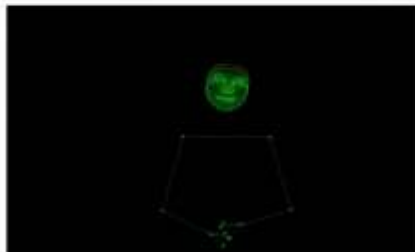
By Michael Erand



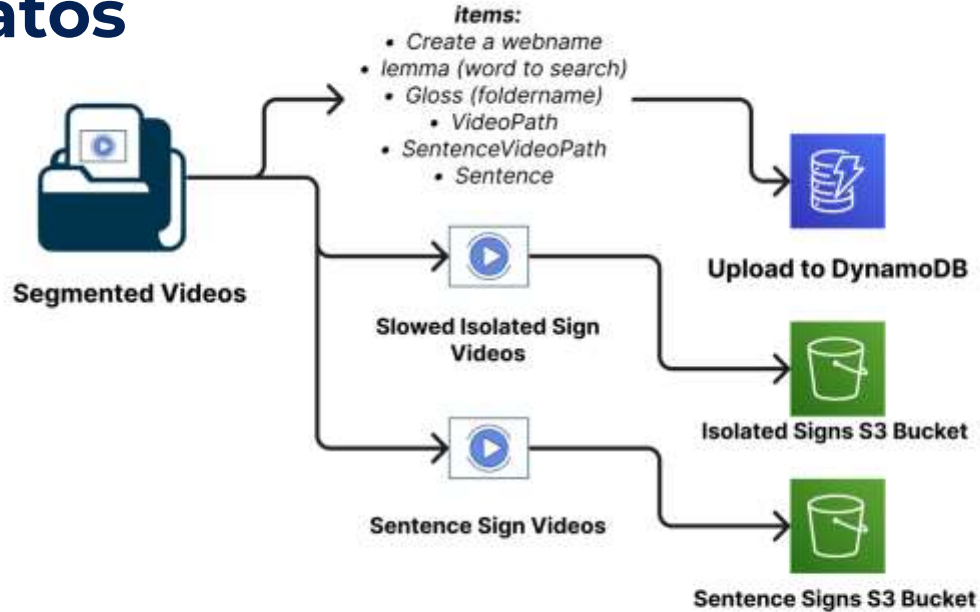
Josh Sussman (left) demonstrates his invention, which claims to "translate" sign language into text and spoken words. (Stephen J. Delaney / AP)



PUCP Diccionario de Lengua de Señas Peruanas

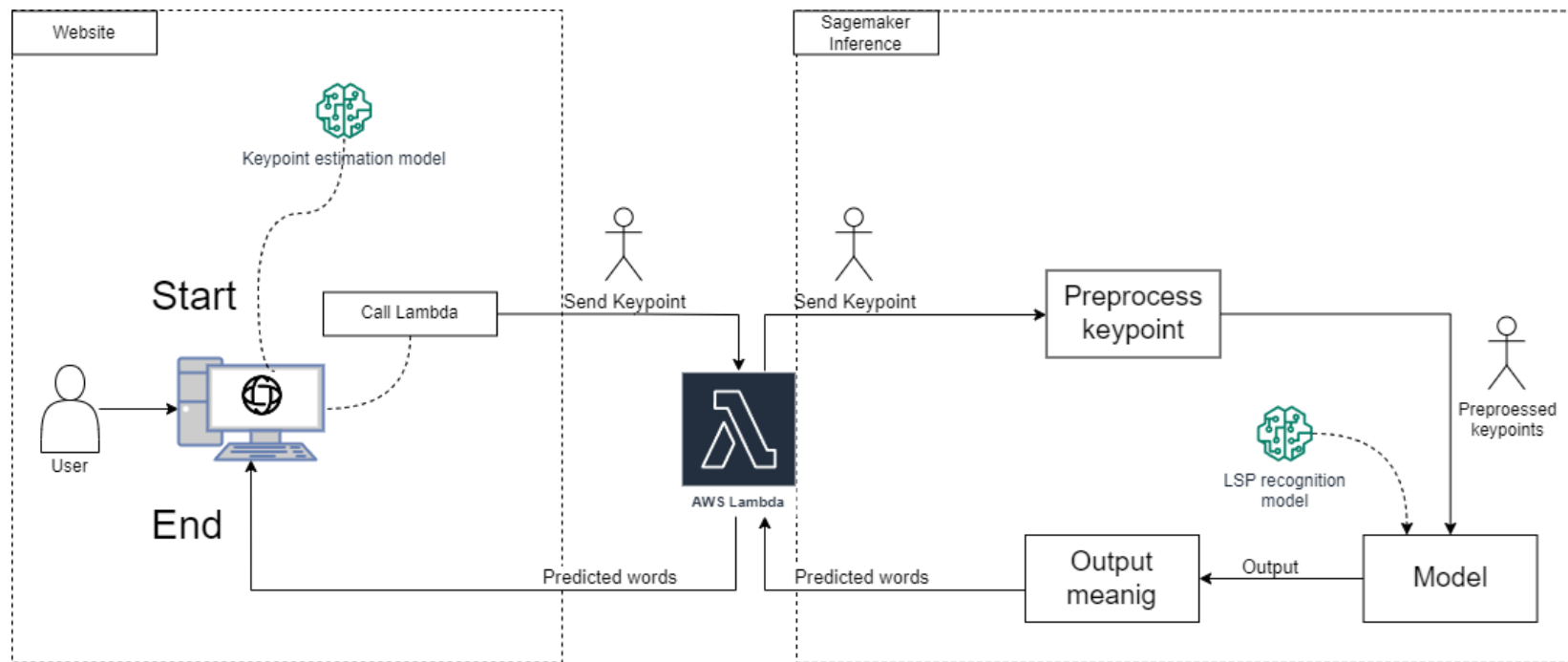


Arquitectura AWS – Almacenamiento de nuestros datos



[Paper](#)

Arquitectura AWS - Búsqueda por seña



Interactuemos con el diccionario Web

Accede al diccionario web diccionariosp.ucp.edu.pe

Realicemos señas y anotemos nuestro accuracy para TOP-1 y TOP-5

- Adentro
- Ayudar
- Dormir
- Sentir
- Bien
- Hola

Peruvian Sign Language Dictionary

- 
1. Equipo
 2. Interesados
 3. Arquitectura Web - AWS
 4. Demo

Agenda

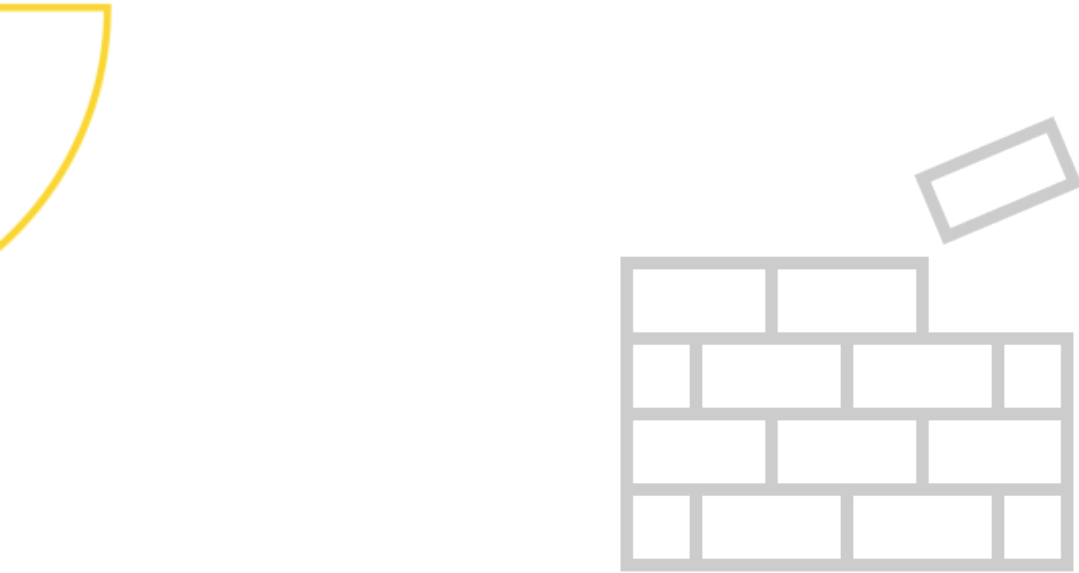
66.67%

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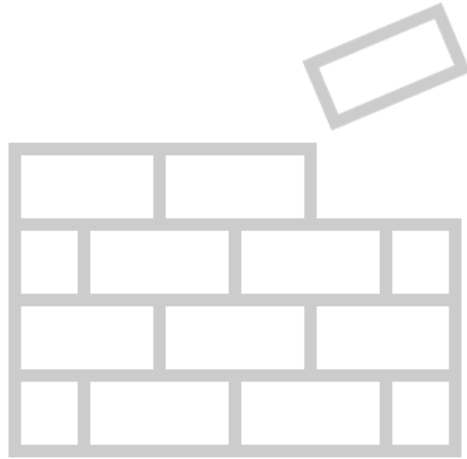
Comparing Incremental Learning Approaches for a Growing Sign Language Dictionary

¿Qué es el incremental learning?

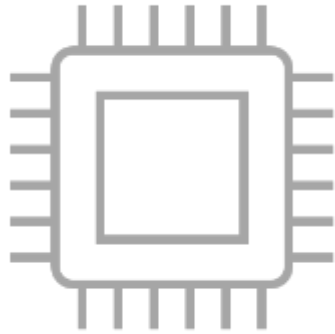


¿Qué es el incremental learning?

Son métodos o técnica que permiten que un modelo incremente sus capacidades continuamente sin realizar entrenamientos desde cero

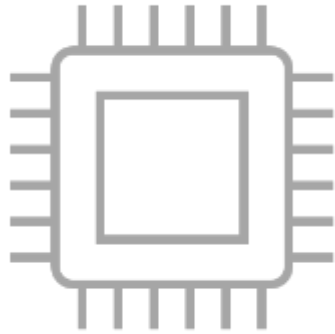


¿Para qué se utiliza el incremental learning?



¿Para qué se utiliza el incremental learning?

- Para reducir la cantidad de datos en durante el entrenamiento
- Para reducir el tiempo de procesamiento
- Para reducir el uso de memoria GPU



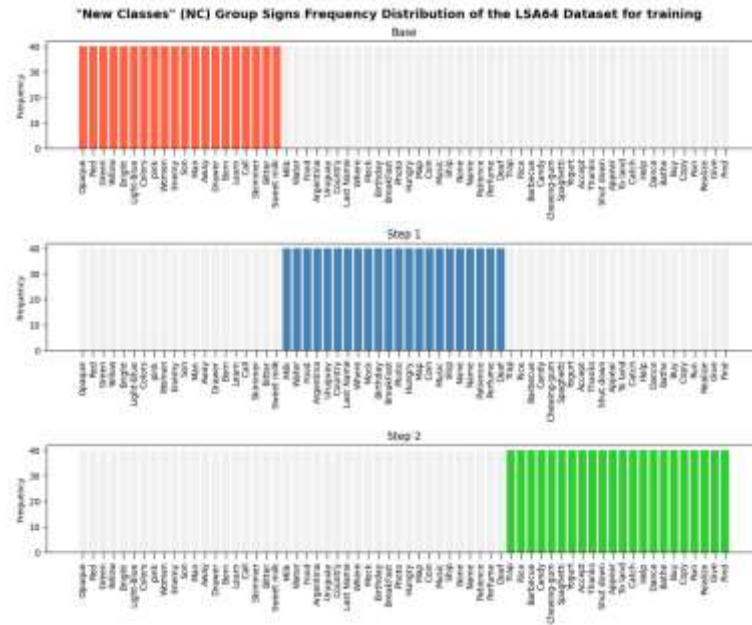
¿Qué tipos de incremental learning hay?



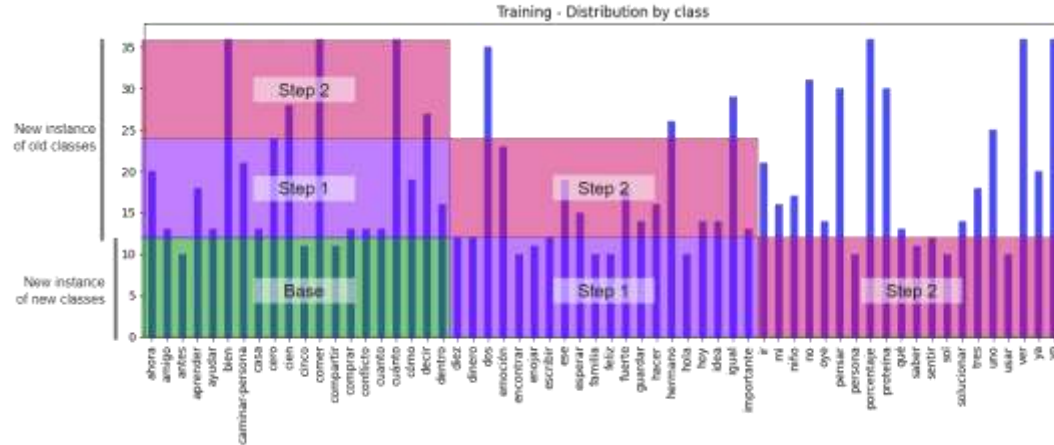
¿Qué tipos de incremental learning hay?

- 
- Class Incremental
 - Task Incremental
 - Domain Incremental

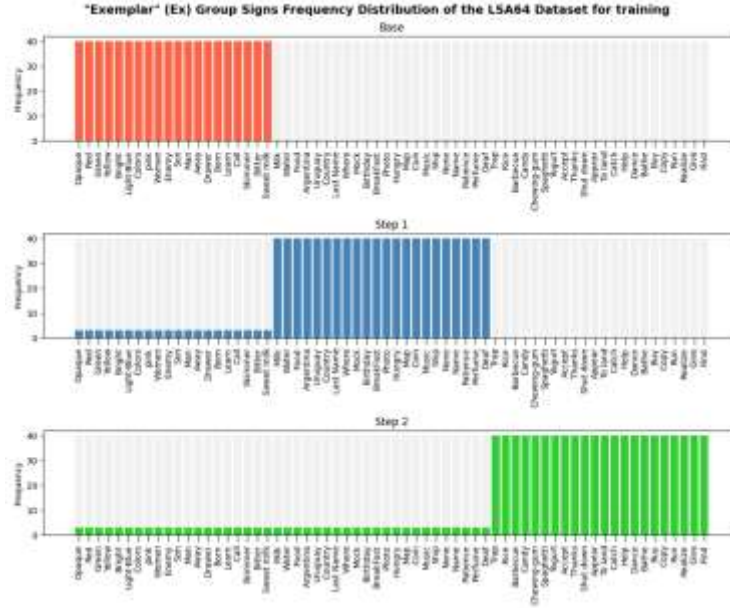
Incremental groups for training - New Classes (NC)

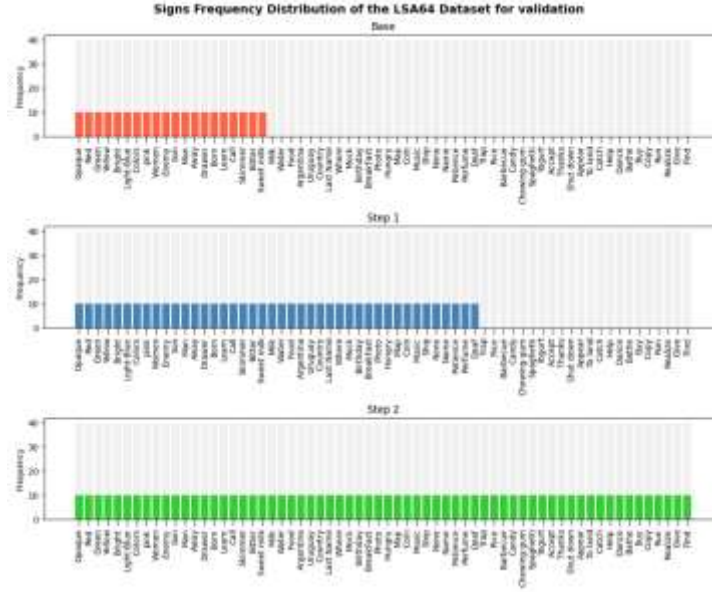


Incremental groups for training - New Instances of New and Old Classes (NINOC)



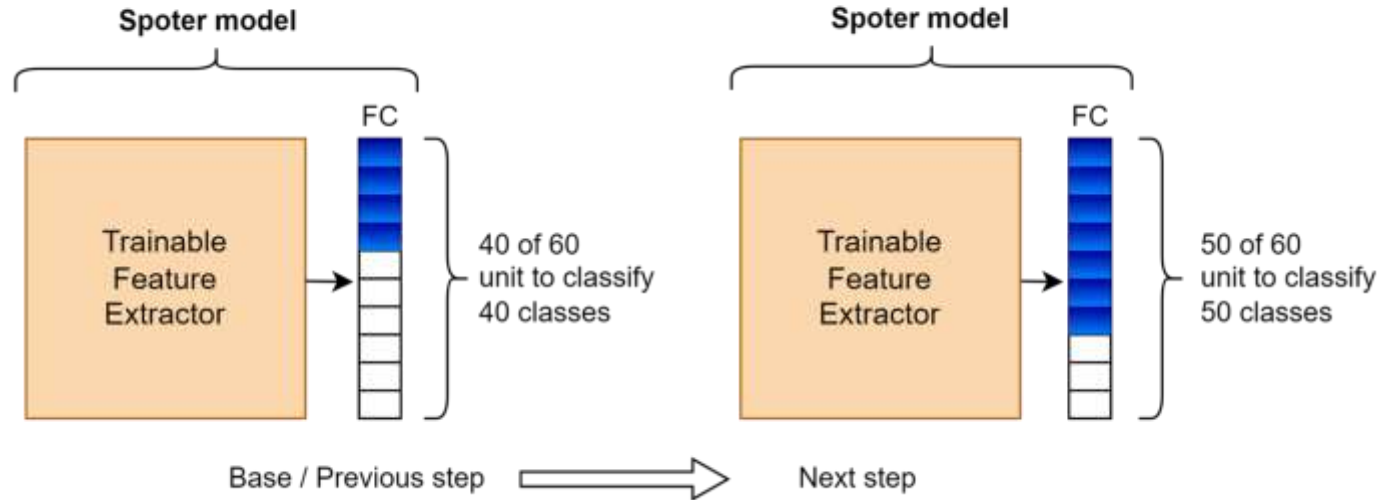
Incremental groups for training - Exemplar (Ex)





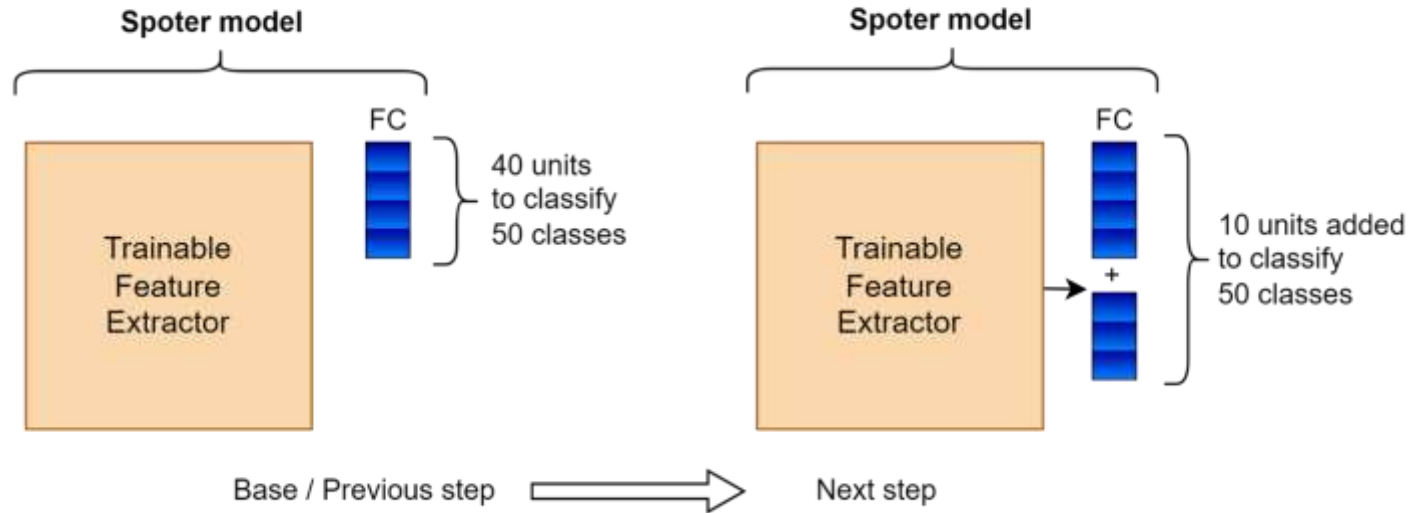
Método 1

Fixed Last Layer (FLL)

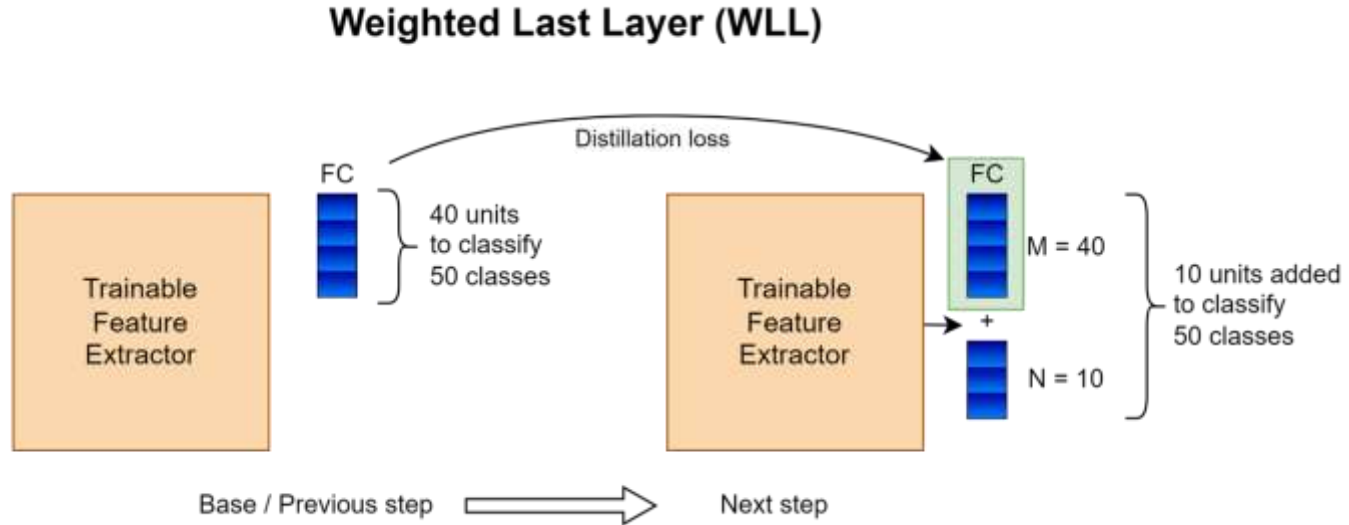


Método 2

Expanded Last Layer (ELL)



Método 3



Método 4

Frozen Feature Extractor - Fixed Last Layer (FLL-Frozen)

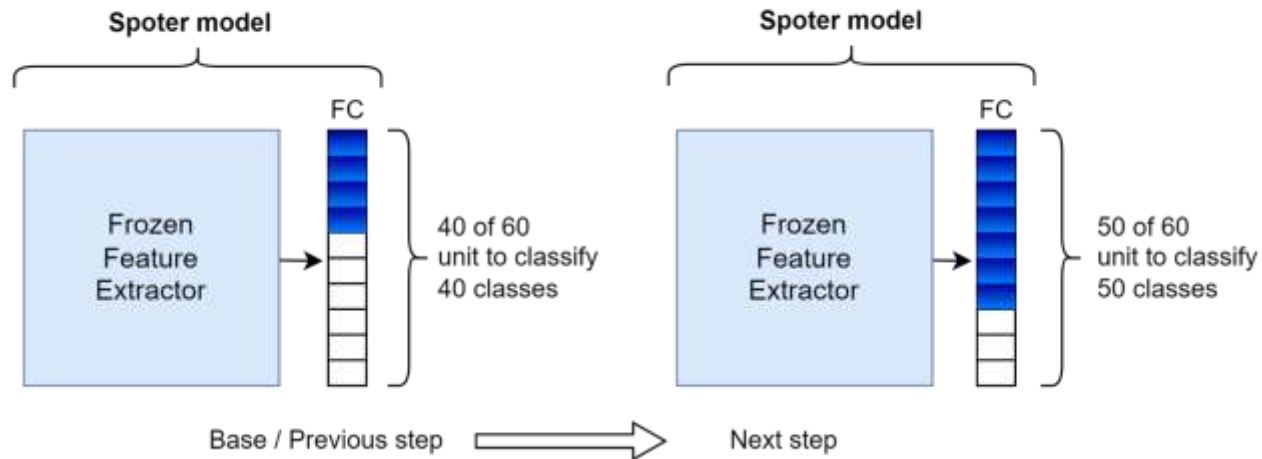


TABLE I

RESULTS OF THE INCREMENT TECHNIQUES USING "20-40-60" STEPS ON THE PUCP-AEC DATASET FOR NC INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Base	20	75.10 $\pm$ 6.20	92.63 $\pm$ 2.05
Base Fixed	20	76.80 $\pm$ 3.86	90.35 $\pm$ 4.02
ELL	20-40	35.31 $\pm$ 0.28	44.36 $\pm$ 3.77
ELL-Frozen	20-40	47.73 $\pm$ 3.67	68.37 $\pm$ 1.98
FL	20-40	33.91 $\pm$ 2.83	42.09 $\pm$ 1.30
FL-Frozen	20-40	44.61 $\pm$ 4.70	70.03 $\pm$ 4.51
WL	20-40	39.83 $\pm$ 1.19	44.93 $\pm$ 3.43
WL-Frozen	20-40	48.86 $\pm$ 4.10	76.27 $\pm$ 1.05
ELL	40-60	23.67 $\pm$ 1.99	28.59 $\pm$ 1.74
ELL-Frozen	40-60	35.80 $\pm$ 2.95	52.65 $\pm$ 4.77
FL	40-60	24.62 $\pm$ 3.13	28.41 $\pm$ 3.94
FL-Frozen	40-60	34.09 $\pm$ 5.42	45.08 $\pm$ 5.37
WL	40-60	26.32 $\pm$ 0.33	28.22 $\pm$ 1.18
WL-Frozen	40-60	35.42 $\pm$ 5.97	64.20 $\pm$ 1.50

TABLE II

RESULTS OF THE INCREMENT TECHNIQUES USING "40-50-60" STEPS ON THE PUCP-AEC DATASET FOR EXEMPLAR INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Expanded	40-50	49.33 $\pm$ 2.56	68.56 $\pm$ 1.98
Expanded frozen	40-50	57.24 $\pm$ 3.76	77.60 $\pm$ 3.18
Fixed	40-50	46.59 $\pm$ 6.01	68.54 $\pm$ 5.55
Fixed frozen	40-50	56.78 $\pm$ 4.57	79.18 $\pm$ 1.78
Weighted	40-50	59.27 $\pm$ 3.74	80.31 $\pm$ 1.42
Weighted frozen	40-50	52.48 $\pm$ 2.20	66.96 $\pm$ 2.84
Expanded	50-60	35.98 $\pm$ 1.18	57.95 $\pm$ 0.57
Expanded frozen	50-60	49.62 $\pm$ 4.19	74.81 $\pm$ 4.59
Fixed	50-60	36.74 $\pm$ 0.87	57.01 $\pm$ 4.19
Fixed frozen	50-60	49.24 $\pm$ 4.03	75.95 $\pm$ 2.92
Weighted	50-60	47.73 $\pm$ 2.60	72.35 $\pm$ 0.33
Weighted frozen	50-60	43.94 $\pm$ 2.00	56.06 $\pm$ 1.43

TABLE III

RESULTS OF THE INCREMENT TECHNIQUES USING "40-50-60" STEPS ON THE PUCP-AEC DATASET FOR NC INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Base	40	65.25 $\pm$ 2.32	80.50 $\pm$ 2.39
Fixed Base	40	65.55 $\pm$ 2.30	82.19 $\pm$ 2.39
Expanded	40-50	37.10 $\pm$ 5.10	46.61 $\pm$ 2.55
Expanded frozen	40-50	57.01 $\pm$ 4.19	78.05 $\pm$ 0.87
Fixed	40-50	30.54 $\pm$ 4.88	47.74 $\pm$ 5.12
Fixed frozen	40-50	57.24 $\pm$ 4.30	79.19 $\pm$ 1.64
Weighted	40-50	55.42 $\pm$ 2.35	77.37 $\pm$ 0.44
Weighted frozen	40-50	52.48 $\pm$ 2.20	66.96 $\pm$ 2.84
Expanded	50-60	26.52 $\pm$ 3.65	36.93 $\pm$ 0.57
Expanded frozen	50-60	50.57 $\pm$ 4.51	70.45 $\pm$ 4.95
Fixed	50-60	19.32 $\pm$ 3.73	34.66 $\pm$ 2.27
Fixed frozen	50-60	49.81 $\pm$ 4.26	73.67 $\pm$ 5.91
Weighted	50-60	45.08 $\pm$ 1.18	68.75 $\pm$ 0.98
Weighted frozen	50-60	43.94 $\pm$ 2.00	55.87 $\pm$ 1.43

TABLE IV

RESULTS OF THE INCREMENT TECHNIQUES USING "20-40-60" STEPS ON THE PUCP-AEC DATASET FOR EXEMPLAR INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Expanded	20-40	47.74 $\pm$ 1.71	67.48 $\pm$ 5.46
Expanded frozen	20-40	48.02 $\pm$ 1.93	72.03 $\pm$ 1.49
Fixed	20-40	45.19 $\pm$ 4.72	67.21 $\pm$ 4.39
Fixed frozen	20-40	45.47 $\pm$ 2.45	72.29 $\pm$ 4.38
Weighted	20-40	53.66 $\pm$ 2.15	77.67 $\pm$ 3.68
Weighted frozen	20-40	49.13 $\pm$ 5.64	76.55 $\pm$ 0.53
Expanded	40-60	35.98 $\pm$ 0.87	57.20 $\pm$ 2.37
Expanded frozen	40-60	36.93 $\pm$ 2.05	58.33 $\pm$ 5.16
Fixed	40-60	37.88 $\pm$ 2.80	58.71 $\pm$ 1.18
Fixed frozen	40-60	35.23 $\pm$ 2.60	50.76 $\pm$ 3.33
Weighted	40-60	39.58 $\pm$ 1.18	68.94 $\pm$ 2.62
Weighted frozen	40-60	35.42 $\pm$ 7.10	61.55 $\pm$ 4.26



[Paper](#)

LSA64 - Argentino

TABLE V
RESULTS OF THE INCREMENT TECHNIQUES USING "20-42-64" STEPS ON
THE LSA64 DATASET FOR NC INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Base	20	98.50 $\pm$ 0.50	100.00 $\pm$ 0.00
Fixed Base	20	99.17 $\pm$ 0.58	100.00 $\pm$ 0.00
Expanded	20-42	52.54 $\pm$ 0.50	78.10 $\pm$ 1.86
Expanded frozen	20-42	66.35 $\pm$ 3.57	95.08 $\pm$ 1.94
Fixed	20-42	53.17 $\pm$ 0.99	77.06 $\pm$ 3.82
Fixed frozen	20-42	69.29 $\pm$ 3.43	93.02 $\pm$ 2.35
Weighted	20-42	75.87 $\pm$ 5.02	99.29 $\pm$ 0.63
Weighted frozen	20-42	81.03 $\pm$ 2.38	99.52 $\pm$ 0.63
Expanded	42-64	33.96 $\pm$ 0.48	44.74 $\pm$ 4.65
Expanded frozen	42-64	43.7 $\pm$ 1.77	80.23 $\pm$ 2.32
Fixed	42-64	34.27 $\pm$ 0.33	47.86 $\pm$ 1.98
Fixed frozen	42-64	43.67 $\pm$ 2.76	83.44 $\pm$ 1.33
Weighted	42-64	40.00 $\pm$ 1.34	85.21 $\pm$ 5.23
Weighted frozen	42-64	59.11 $\pm$ 1.52	86.61 $\pm$ 18.45

TABLE VI
RESULTS OF THE INCREMENT TECHNIQUES USING "20-42-64" STEPS ON
THE LSA64 DATASET FOR EXEMPLAR INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Expanded	20-42	93.10 $\pm$ 1.67	99.05 $\pm$ 0.24
Expanded frozen	20-42	74.68 $\pm$ 2.82	99.44 $\pm$ 0.36
Fixed	20-42	93.73 $\pm$ 1.62	99.05 $\pm$ 0.24
Fixed frozen	20-42	77.14 $\pm$ 2.27	99.13 $\pm$ 0.73
Weighted	20-42	96.98 $\pm$ 1.07	100.00 $\pm$ 0.00
Weighted frozen	20-42	89.21 $\pm$ 0.99	99.68 $\pm$ 0.55
Expanded	42-64	90.89 $\pm$ 2.19	98.75 $\pm$ 0.72
Expanded frozen	42-64	56.51 $\pm$ 1.58	94.22 $\pm$ 2.43
Fixed	42-64	91.51 $\pm$ 0.63	98.96 $\pm$ 0.09
Fixed frozen	42-64	55.52 $\pm$ 0.59	94.48 $\pm$ 3.00
Weighted	42-64	95.94 $\pm$ 0.41	99.95 $\pm$ 0.09
Weighted frozen	42-64	74.64 $\pm$ 4.38	98.44 $\pm$ 0.81

TABLE VII
RESULTS OF THE INCREMENT TECHNIQUES USING "44-54-64" STEPS ON
THE LSA64 DATASET FOR NC INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Base	44	99.09 $\pm$ 0.45	100.00 $\pm$ 0.0
Fixed Base	44	99.17 $\pm$ 0.26	99.92 $\pm$ 0.13
Expanded	44-54	31.54 $\pm$ 1.51	86.11 $\pm$ 1.82
Expanded frozen	44-54	89.38 $\pm$ 1.18	99.94 $\pm$ 0.11
Fixed	44-54	28.02 $\pm$ 2.08	85.25 $\pm$ 2.65
Fixed frozen	44-54	89.8 $\pm$ 1.21	99.88 $\pm$ 0.11
Weighted	44-54	96.54 $\pm$ 1.02	99.88 $\pm$ 0.21
Weighted frozen	44-64	97.59 $\pm$ 0.49	99.88 $\pm$ 0.11
Expanded	54-64	22.50 $\pm$ 0.56	70.57 $\pm$ 0.39
Expanded frozen	54-64	78.85 $\pm$ 5.71	99.48 $\pm$ 0.24
Fixed	54-64	23.96 $\pm$ 3.58	71.93 $\pm$ 3.40
Fixed frozen	54-64	79.69 $\pm$ 4.25	99.32 $\pm$ 0.18
Weighted	54-64	80.99 $\pm$ 2.90	99.53 $\pm$ 0.54
Weighted frozen	54-64	82.24 $\pm$ 0.33	94.43 $\pm$ 0.63

TABLE VIII
RESULTS OF THE INCREMENT TECHNIQUES USING "44-54-64" STEPS ON
THE LSA64 DATASET FOR EXEMPLAR INCREMENTAL GROUPS

Increment technique	Increment step	acc top 1 (%)	acc top 5 (%)
Expanded	44-54	95.00 $\pm$ 0.49	99.94 $\pm$ 0.11
Expanded frozen	44-54	93.02 $\pm$ 1.20	100.00 $\pm$ 0.00
Fixed	44-54	94.81 $\pm$ 0.49	99.81 $\pm$ 0.00
Fixed frozen	44-54	94.20 $\pm$ 1.09	100.00 $\pm$ 0.00
Weighted	44-54	98.09 $\pm$ 0.11	100.00 $\pm$ 0.00
Weighted frozen	44-54	97.04 $\pm$ 0.19	99.94 $\pm$ 0.11
Expanded	54-64	91.48 $\pm$ 0.33	99.61 $\pm$ 0.11
Expanded frozen	54-64	87.60 $\pm$ 4.28	99.69 $\pm$ 0.16
Fixed	54-64	92.97 $\pm$ 1.28	99.43 $\pm$ 0.18
Fixed frozen	54-64	88.80 $\pm$ 5.70	99.79 $\pm$ 0.09
Weighted	54-64	97.66 $\pm$ 0.41	100.00 $\pm$ 0.00
Weighted frozen	54-64	94.27 $\pm$ 1.19	99.58 $\pm$ 0.09



[Code](#)

Comparing Incremental Learning Approaches for a Growing Sign Language Dictionary

- 
1. Conceptos
 2. Configuración
 3. Métodos
 4. Resultados
- 

Agenda

75%

1. Introducción básica del trabajo con Lengua de seña
2. PERUSIL: a Sign Language Annotation Framework
3. Machine Learning used to Measure Sign Language Similarity
4. Impact of Pose Estimation on the Sign Language Recognition Task
5. Peruvian Sign Language Dictionary
6. Comparing Incremental Learning Approaches for a Growing Sign Language Dictionary
- 7. AI for Sign Language Data Interpolation**
8. Iconicity in Sign Language



AI for Sign Language Data Interpolation

¿Por qué es importante la interpolación en SL?



¿Por qué es importante la interpolación en SL?

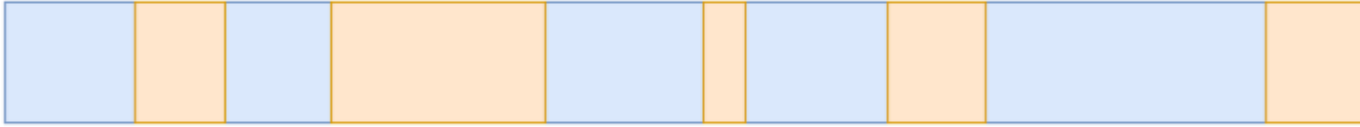
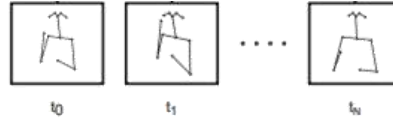
Para reducir el impacto de los missing values en los modelos

Para poder hacer avatars o representaciones 2D/3D

Para agregar más frames al video de señas



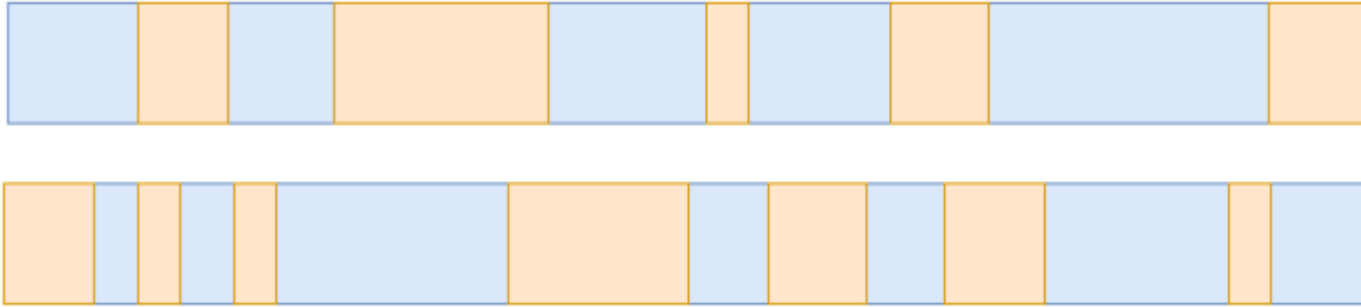
Missing Values



Serie de de Keypoints de un Video

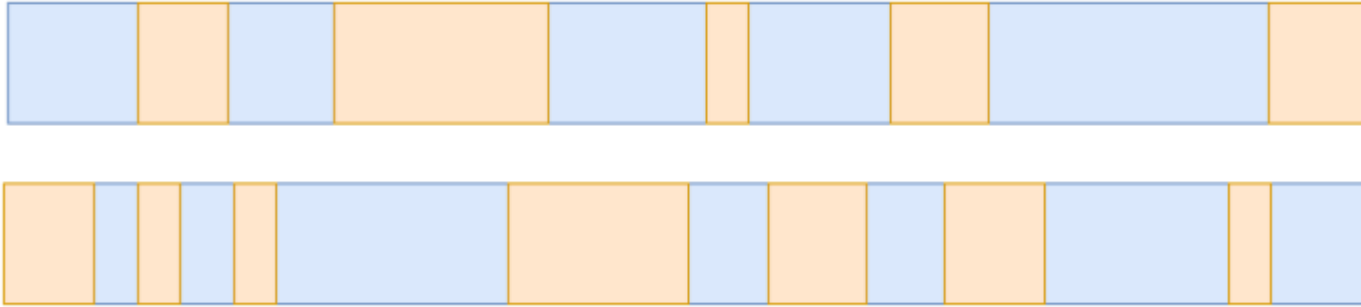
Missing Values

¿Cómo simular cuantos bloques hay?



Missing Values

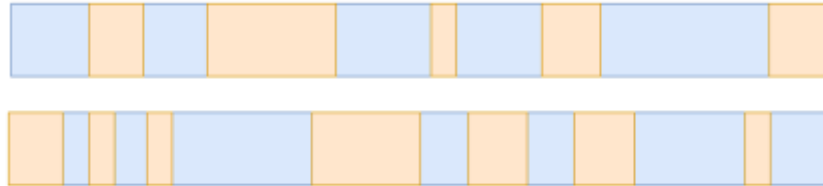
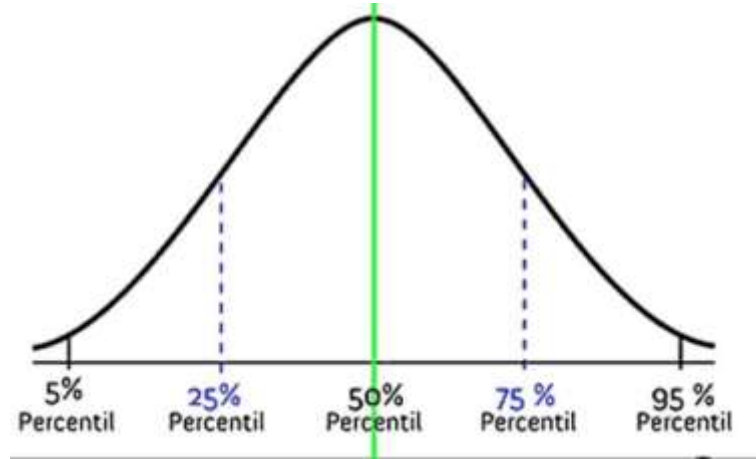
¿Cómo simular el tamaño de los bloques?



Missing Values

¿Cómo simular
cuantos bloques
hay?

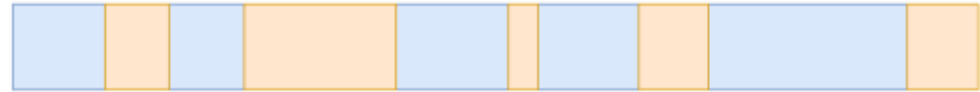
¿Cómo simular el
tamaño de los
bloques?



Simular Missing Values

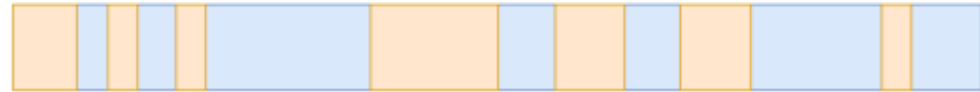
¿Cómo simular
cuantos bloques
hay?

$[2, 5]$



¿Cómo simular el
tamaño de los
bloques?

$[1, 4]$



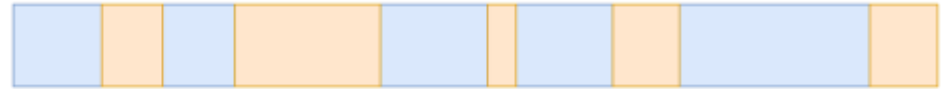
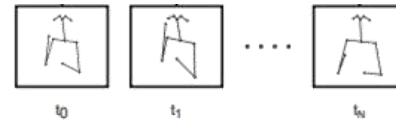
Tipos simples de interpolación para missing values

Repetir el anterior/posterior

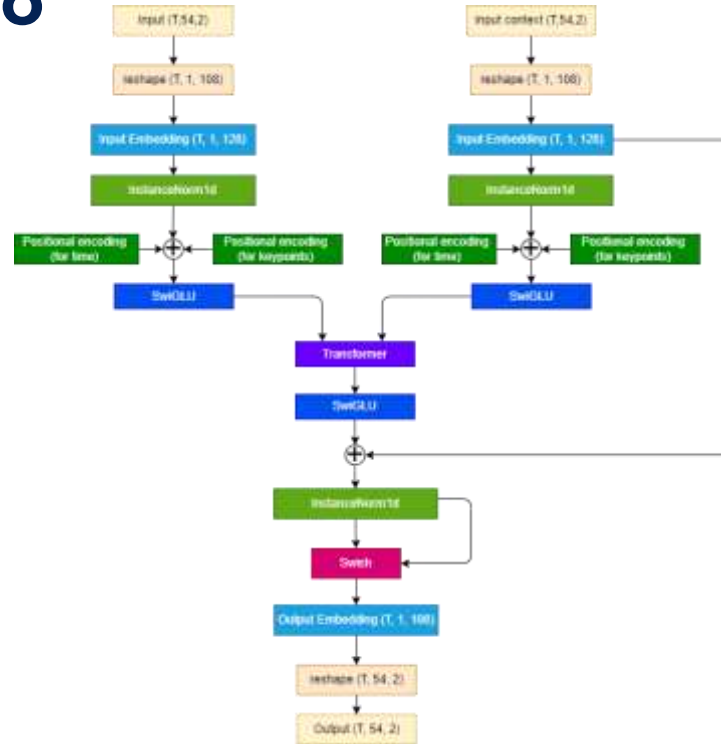
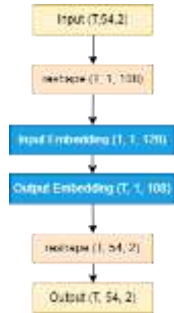
Interpolación cúbica

Interpolación Splines

Interpolación Spline-cúbica



Arquitectura del modelo



Máscaras

src_mask

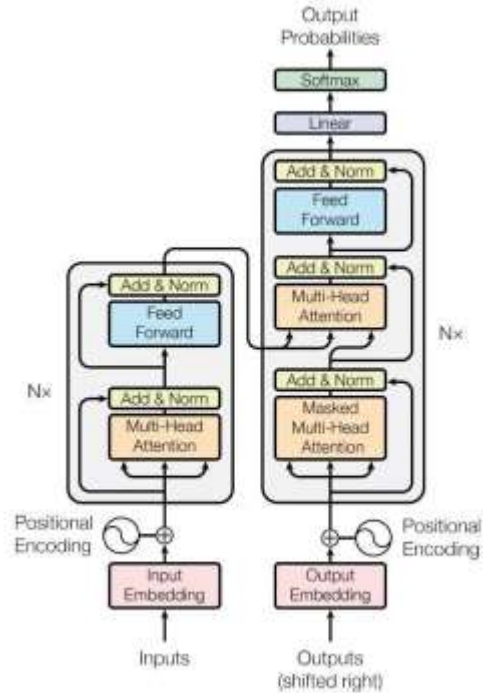
tgt_mask

src_key_padding_mask

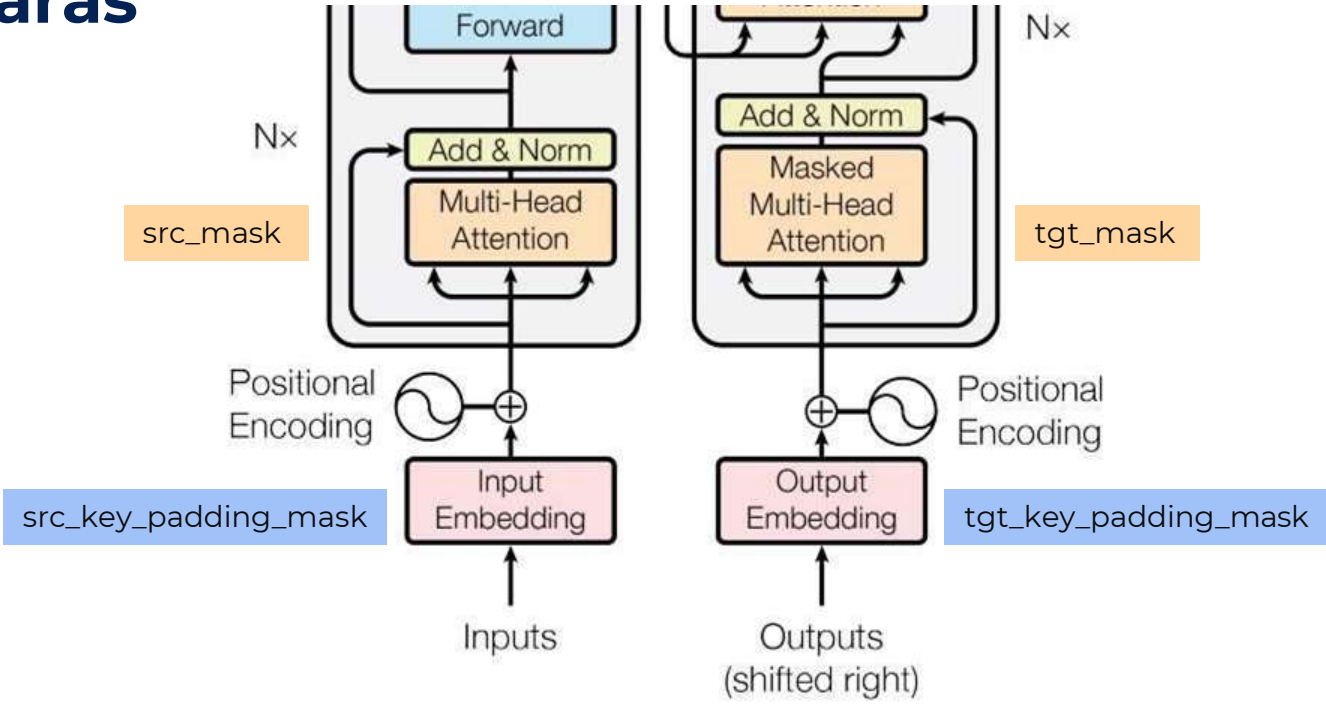
tgt_key_padding_mask

memory_mask

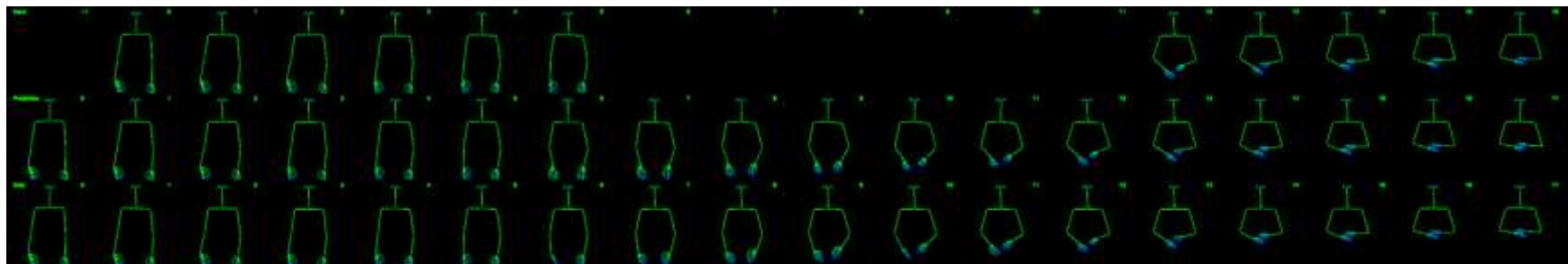
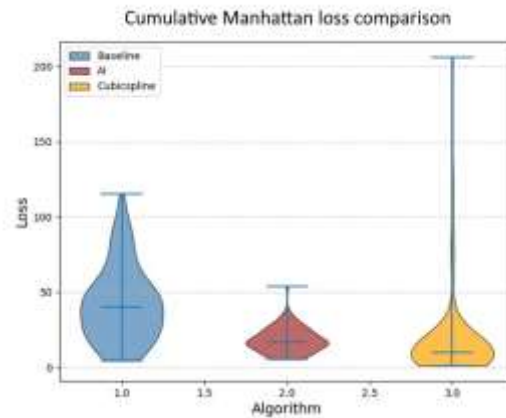
memory_key_padding_mask



Máscaras



Resultados



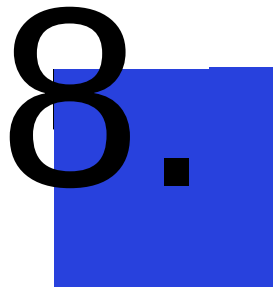
AI for Sign Language Data Interpolation

- 
1. Conceptos
 2. Missing values
 3. Arquitectura
 4. Resultados

Agenda

85.42%

1. Introducción básica del trabajo con Lengua de seña
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7. AI for Sign Language Data Interpolation
- 8. Iconicity in Sign Language**



AI2050

Iconicity in Sign Language

Collect Dataset

Review SL similarity list
Review (Ortega et al, 2019) for more iconic list

sign/gestures
videos



images



List of signs (videos) - images

Collect Dataset

Review SL similarity list
Review (Ortega et al, 2019) for more iconic list

sign/gestures
videos



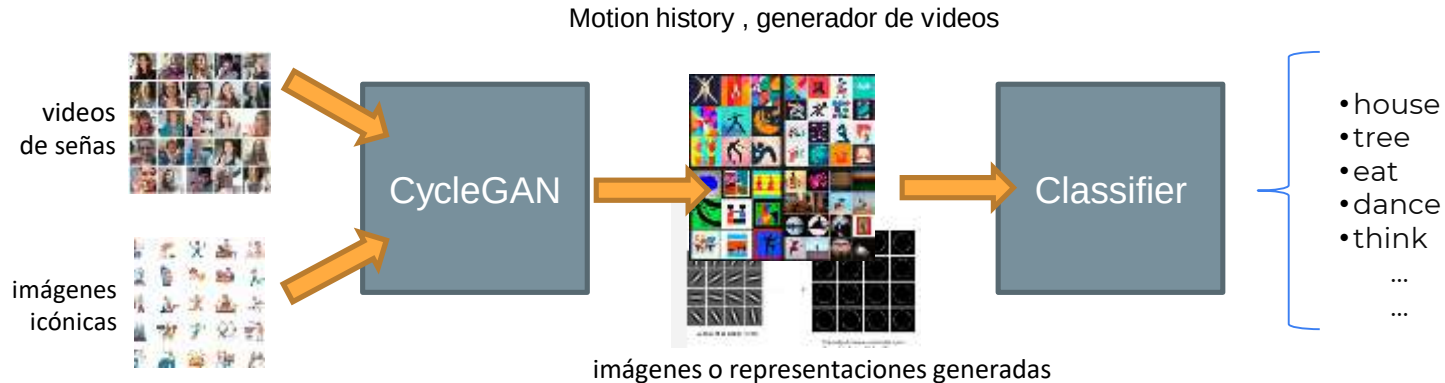
images



List of signs (videos) - images

Building the AI Models

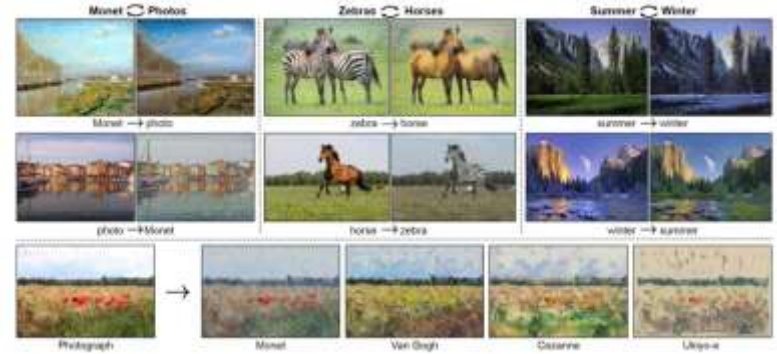
1. Train a generative GAN to transfer style between videos and associated images (cycleGAN)
2. Train a classifier to recognize the image representing the sign



Training the CycleGAN

CycleGAN is a model that Will transfer style from one image to another.

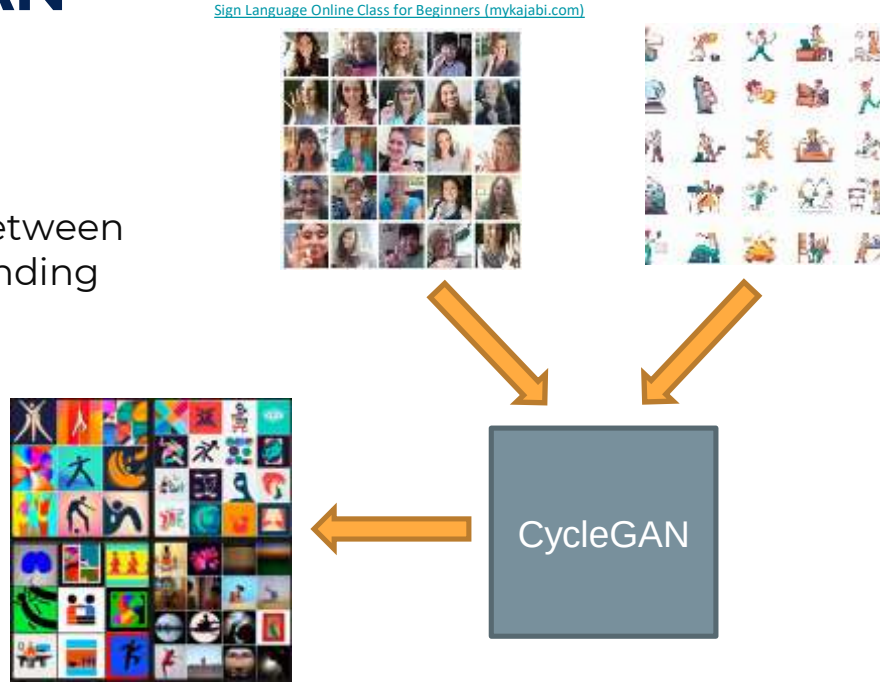
For example, we see the picture of a horse and we want to convert it in a zebra given a second input related to the domain zebra.



<https://www.youtube.com/watch?v=-8hfnlxEpn4>

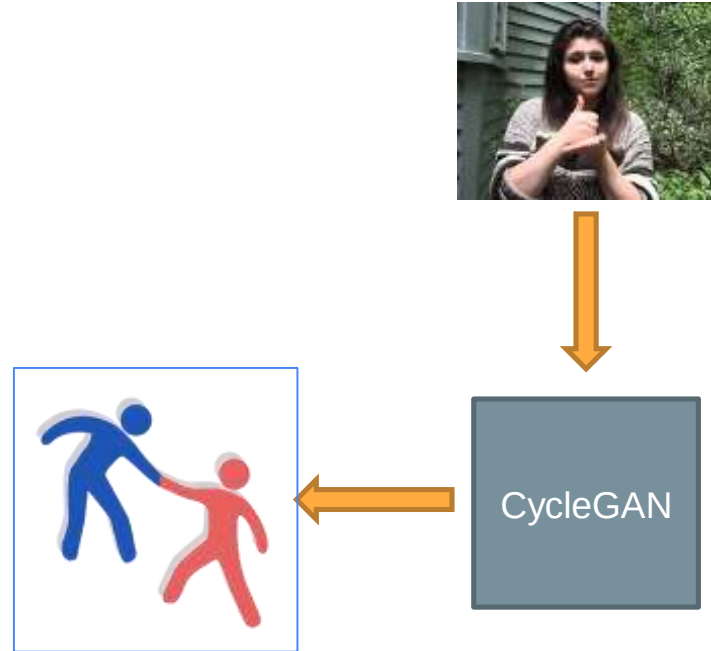
Training the CycleGAN

In our Project, at training time, the cycleGAN model sees the match between the video of a sign and its corresponding image



Generating time CycleGAN model

At generation time, the CycleGAN model receives one video at a time to generate an image corresponding to that sign according to the patterns that it previously learned



Training the Classifier

The classifier should be able to recognize either “abstract” input as well as well defined pictures

The classes should include signs seen and not seen in the cycleGAN



Object/Action
Classifier

- house
- tree
- eat
- dance
- think
- ...
- ...

Referencias

Gramática de la LSP

https://youtu.be/ELH8FBg4byg?si=vCBOjbV-tZy8_e7m

Segmentación

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9413817>

Iconicity in Sign Language

1. Datasets
2. Arquitectura
3. Entrenamiento



Agenda

100%

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¡Gracias!



en American Sign Language

¡Gracias!



[Website personal](#)